National University of Computer and Emerging Sciences



Dataset Report:
Sentence-Role Classification on PubMed RCT

Fine-Tuned Biomedical Transformers vs. Classical Machine Learning

F26-07

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Semester: 5

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1. Dataset Link and Description

Source: https://github.com/Franck-Dernoncourt/pubmed-rct

PubMed RCT is a benchmark corpus of abstracts of randomized controlled trials (RCTs) taken from PubMed. Every abstract is split into sentences, and every sentence carries a label naming the *rhetorical role* it plays in the abstract. There are five roles:

- **BACKGROUND:** context and motivation for the study
- **OBJECTIVE:** the aim or hypothesis
- **METHODS:** study design, participants, interventions
- **RESULTS:** findings, measurements, statistics
- **CONCLUSIONS:** the take-home message

The corpus comes in two sizes (about 20k and 200k abstracts). We use the **20k version**, which is large enough for meaningful model comparison yet small enough to fine-tune transformers within a semester on free GPU resources.

In the raw files, each abstract begins with a line `###<abstract_id>`, followed by one line per sentence in the form `LABEL<TAB>sentence`. For example, abstract 20497432 (thyroid ophthalmopathy trial) opens with a BACKGROUND sentence stating the study aim, moves through METHODS and RESULTS sentences with patient counts and measurements, and ends with two CONCLUSIONS sentences.

Learning problem: supervised, sentence-level, 5-class text classification. The **input (X)** is the sentence text plus structural features such as its position in the abstract, and the **target (y)** is the role label.

2. Data Collection Method

The dataset was not scraped or assembled by us. We **cloned the public GitHub repository** and worked with the **20k training split**, which is the largest portion and the one used for exploration. The validation and test splits were left untouched so they can serve for unbiased model evaluation later. This also keeps our results comparable with published work on the same benchmark.

3. Preprocessing Steps

The raw text file was converted into a structured pandas DataFrame through the following steps:

1. **Cleaning:** removed empty lines and stray whitespace, and stripped the # characters from the abstract header lines.
2. **Parsing:** extracted the abstract ID, label, and sentence text from each line. Sentences were grouped under the abstract they belong to.
3. **Feature engineering:** added structural attributes for every sentence:

- `line_number`: index of the sentence within its abstract
- `total_lines`: number of sentences in that abstract
- `relative_position`: `line_number / total_lines`, a value between 0 (start) and 1 (end)
- `line_length`: sentence length in characters

The final DataFrame has seven columns: `abstract_id`, `label`, `text`, `line_number`, `total_lines`, `line_length`, `relative_position`.

We deliberately kept the cleaning light. Numbers, percentages, and punctuation were **not** removed, because they are strong cues for RESULTS sentences, and stopwords were kept because words like "was", "will", and "we" carry tense information that separates METHODS from RESULTS.

4. Brief Exploratory Data Analysis

All plots referenced below (Figures 1 to 4) are not repeated in this report. They are generated in the accompanying Jupyter notebook (.ipynb), where they can be viewed alongside the code that produced them.

4.1 Class distribution

The training set holds **180,040 sentences** across five classes, and they are clearly imbalanced.

Label	Sentences	Share
METHODS	59,353	33.0%
RESULTS	57,953	32.2%
CONCLUSIONS	27,168	15.1%
BACKGROUND	21,727	12.1%
OBJECTIVE	13,839	7.7%

METHODS and RESULTS together make up about two thirds of the data, while OBJECTIVE is the minority class at under 8%. A model that favors the big classes can still reach high accuracy, so **macro-F1 and per-class metrics** must be reported, and class weighting is worth testing.

Figure 1: Class label distribution (bar chart). See the project notebook.

4.2 Label position within the abstract

The density plot of relative position shows that abstracts follow a strongly ordered structure:

- **OBJECTIVE** spikes at the very start (position near 0) with a smaller bump around 0.08.
- **BACKGROUND** is concentrated in the first 20% of the abstract.
- **METHODS** peaks between roughly 0.1 and 0.45.

- **RESULTS** dominates the middle-to-late region, about 0.45 to 0.8.
- **CONCLUSIONS** forms a sharp peak near 0.9.

The classes are largely separated along this axis, which makes **position a very strong predictive feature**. There is also a small BACKGROUND bump at the very end of abstracts. This likely reflects odd cases such as trial registration numbers labeled BACKGROUND (e.g., `ACTRN12612000642886 .` in our sample) and points to some label noise.

Figure 2: Density of labels by relative position. See the project notebook.

4.3 Abstract length

Most abstracts contain 8 to 16 sentences, with the mode at 11 to 12 sentences (over 4,100 abstracts in that bin). The distribution is slightly right-skewed, with a short tail up to about 30 sentences. The dataset therefore covers roughly 15,000 abstracts with a fairly consistent size, and relative position is computed on a stable denominator.

Figure 3: Distribution of abstract lengths (sentences per abstract). See the project notebook.

4.4 Sentence length by label

Median sentence length is similar across classes: OBJECTIVE 164 characters, RESULTS 142, CONCLUSIONS 140, BACKGROUND 132, and METHODS 131. OBJECTIVE sentences are slightly longer and more uniform, while RESULTS and METHODS show many long outliers (up to about 1,450 characters) from sentences packed with measurements. Length is therefore a weak but not useless feature. It will not separate classes on its own, but it adds mild signal alongside text and position.

Figure 4: Sentence length by label (box plot). See the project notebook.

4.5 Key takeaways

- The five classes are imbalanced, so **macro-F1 is the primary metric**.
- Position is the strongest non-text signal, and length is secondary.
- OBJECTIVE is the smallest class, overlaps with BACKGROUND at the start of abstracts, and is expected to be the hardest to predict.
- Some label noise exists and should be acknowledged as a ceiling on achievable accuracy.

5. Final Dataset Summary

Overview

Property	Value
Source	PubMed RCT (GitHub: FranckDernoncourt/pubmed-rct)
Version / split used	20k, training set

Property	Value
Unit of data	One sentence from a medical abstract
Size	180,040 sentences from about 15,000 abstracts
Target label	label, 5 classes
Text feature	text
Structural features	line_number, total_lines, relative_position, line_length
Main challenge	Class imbalance and BACKGROUND/OBJECTIVE overlap

Our aim

We aim to build a model that reads a single sentence from a biomedical abstract and predicts its role: Background, Objective, Methods, Results, or Conclusions. Our central question is how much a fine-tuned biomedical transformer (such as PubMedBERT or BioBERT) improves over a classical machine learning pipeline (such as TF-IDF with Logistic Regression or SVM).

Why this dataset fits the project

1. **Clean supervised setup:** every sentence has a ground-truth label, so no manual annotation is needed.
2. **Right scale:** 180k sentences give enough data to train both classical models and fine-tune transformers without excessive compute.
3. **Rich signals for both model families:** the sentence text suits transformers and TF-IDF, while position and length features give classical models a strong structural boost. This lets us test whether transformers still hold an advantage once position is available to everyone.
4. **Meaningful evaluation:** the class imbalance and the BACKGROUND/OBJECTIVE confusion make macro-F1 and per-class analysis informative rather than trivial.
5. **Real-world value:** accurate sentence-role prediction supports automatic structuring of unstructured abstracts, targeted literature search, and evidence extraction for systematic reviews.

With the data parsed, cleaned, and explored, the project moves next to baseline classical models, transformer fine-tuning, and a side-by-side comparison on the held-out validation and test sets.